

b. What is meant by double refraction? Explain Quarter waveplate and Half waveplate with necessary diagrams.

32.a.i. Deduce the expression for the ratio of spontaneous emission rate and stimulated emission rate. (4 Marks)

ii. With neat diagram, explain the construction and working of Nd: YAG laser. (8 Marks)

(OR)

b.i. Based on refractive index profile, explain the classification of optical fibers. (8 Marks)

ii. Discuss the application of optical fiber as pressure sensor. (4 Marks)

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Reg. No.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2019

First / Second Semester

18PYB101J – PHYSICS: ELECTROMAGNETIC THEORY, QUANTUM MECHANICS, WAVES AND OPTICS

(For the candidates admitted during the academic year 2018-2019 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- _____ is defined as the number of magnetic lines of force passing through an unit area of crosssection.
(A) Magnetic flux density (B) Magnetic flux
(C) Magnetic intensity (D) Magnetic permeability
- The vector field whose divergence is zero is called _____.
(A) Irrotational (B) Rotational
(C) Solenoidal (D) Tensor
- The continuity equation is a combination of which of the two laws?
(A) Ohm's law and Gauss law (B) Ampere law and Gauss law
(C) Ohm's law and Ampere law (D) Maxwell's law and Faraday law
- The orientation polarization _____ with increase in temperature.
(A) Remains constant (B) Doubles
(C) Increases (D) Decreases
- The magnetic moment is measured in terms of Bohr magneton (μ_B) and the value of $1\mu_B$ is equal to _____.
(A) $9.27 \times 10^{-24} \text{ A/m}^2$ (B) $9.48 \times 10^{-24} \text{ A/m}^2$
(C) $9.27 \times 10^{-4} \text{ A/m}^2$ (D) $9.48 \times 10^{-4} \text{ A/m}^2$
- In soft magnetic materials, the nature of hysteresis loop is _____.
(A) Very broad (B) Circle
(C) Very steep (D) Straight line
- A tiny movable magnetized cylindrical volume in a thin magnetic material is called _____.
(A) Garnet (B) Magnetoplumbite
(C) Ferrite (D) Magnetic Bubble
- Two ferromagnets which are separated by a few atomic layer of insulator is known as _____.
(A) Hexaferrite (B) Tunnel Magnetoresistance
(C) Multiferroics (D) Colossal Magnetoresistance

9. A variable quantity which characterizes de-Broglie waves is known as _____.
 (A) Wave function (B) Photon
 (C) Phonon (D) Neutron
10. A wave function satisfying the condition _____ is called normalized wave function.
 (A) $\iiint \psi^2 d\tau = 1$ (B) $\iiint \psi^2 d\tau = 0$
 (C) $\iiint \psi^2 d\tau = 0$ (D) $|\psi|^2 = 0$
11. Classical mechanics failed to explain the spectrum of _____ atoms.
 (A) Nitrogen (B) Oxygen
 (C) Hydrogen (D) Potassium
12. Blackbody radiation depends on _____.
 (A) Shape and sizes of black body (B) Volume
 (C) Stress and strain (D) Temperature
13. The wave nature of light is demonstrated by _____ effect.
 (A) Ray tracing (B) Diffraction
 (C) Photosynthesis (D) Photovoltaic
14. In Fresnel diffraction, the source of light is kept at _____ distance from the aperture.
 (A) Finite (B) Zero
 (C) Infinite (D) Small
15. When the ordinary and extraordinary rays travels along the optical axis, then they travel with _____.
 (A) Same velocity (B) Different velocity
 (C) Negative velocity (D) Positive and Negative velocity
16. If the original beam after passing through quarter waveplate is extinguished twice in each rotation, when studied by nicol prism, it is _____.
 (A) Plane polarized (B) Elliptically polarized
 (C) Unpolarized light (D) Circularly polarized
17. A pair of mirrors placed on either side of the active medium is known as _____.
 (A) Optical resonator (B) Cavity
 (C) Shielding (D) Switch
18. In _____ mode, the oxygen atom oscillates along the axis of the molecule simultaneously departing or approaching the carbon atom which is stationary.
 (A) Bending (B) Symmetric stretching
 (C) Asymmetric stretching (D) Wagging
19. In single mode optical fiber, the (V) number is less than _____.
 (A) 0.5 (B) 0.25
 (C) 1.4 (D) 2.4
20. In a _____ fiber, the refractive index changes in a stepwise fashion from the centre of fiber to cladding.
 (A) Glass (B) Step index
 (C) Plastic (D) Graded index

PART – B (5 × 4 = 20 Marks)
 Answer ANY FIVE Questions

21. Derive Laplace equation for Electrostatic potential.
22. Compare four properties of Hard and Soft magnetic materials.
23. Explain photoelectric effect with necessary diagram.
24. X-ray of wavelength 0.1 nm are scattered at such an angle that the recoil electron has maximum kinetic energy. Calculate the energy of recoil electron.
 [Given : $\Delta\lambda = 0.00486 \text{ nm}$, when X-ray's are scattered at 180°].
25. State and derive Brewster's law for polarization of light.
26. If 't' is the minimum thickness of a quarter waveplate needed to convert plane polarized light of wavelength 480 nm into circularly polarized light. Find the corresponding thickness of quarter waveplate for wavelength 600 nm.
27. With neat diagram, explain the working of Homojunction laser.

PART – C (5 × 12 = 60 Marks)
 Answer ALL Questions

28. a. Obtain Maxwell's equation of electromagnetism from the fundamental laws of electricity and magnetism.
 (OR)
 b. What is meant by Internal electric field? Derive an expression for the internal field in dielectrics.
29. a. What are ferrites? Explain the structure of ferrite with diagram. Also, write two applications of ferrites.
 (OR)
 b.i. An Iron rod is subjected to a magnetizing field of 1200 A/m. The susceptibility of Iron is 599. Find the permeability and the magnetic flux per unit area produced. (4 Marks)
 ii. What are multiferroic materials? Explain the types and mention two applications of multiferroic materials. (8 Marks)
30. a. Derive Schrodinger's time independent and dependent wave equation.
 (OR)
 b. Derive Schrodinger wave equation to find ground state wavefunction and energy for quantum harmonic oscillator.
31. a. Define Fraunhofer diffraction. Explain Fraunhofer diffraction at single slit and determine the width of central maxima.

(OR)